



RKDF UNIVERSITY, BHOPAL

B.Sc.(Maths)

SCHEME

Semester – I

No	Subject Code	Subject Type	Subject Title	Marks Allotted						
				Assignment Marks		Theory Marks		Practical Marks		Total Marks
				Max	Min	Max	Min	Max	Min	
1	FC-101/1	Core	English Language	10	4	40	14	-	-	50
2	FC-101/2	Core	Development of Entrepreneurship	10	4	40	14	-	-	50
3	BMM 111	Core	Algebra and Trigonometry	20	8	80	27	-	-	100
4	BCH 111	Core	Based on Inorganic, Organic And Physical Chemistry	20	8	80	27	50	17	150
5	BPY 111	Core	Mechanics	20	8	80	27	50	17	150
Total				80	32	320	109	100	34	500



RKDF UNIVERSITY, BHOPAL

B.Sc.(Maths)

B.C.A./B.A./ B.Com./B.Sc

Semester – I

Course	Subject	Subject Code
B.Sc. (PCM)	English Language	FC-101/1

Unit-I

1. Amalkanti : Nirendranath Chakrabarti
2. Sita : Toru Dutt
3. Tryst with Destiny : Jawaharlal Nehru
4. Delhi in 1857 : Mirza Ghalib
5. Preface to the Mahabharata : C. Rajagopalachari
6. Where the Mind is Without Fear : Rabindranath Tagore
7. A Song of Kabir : Translated by Tagore
8. Satyagraha : M.K. Gandhi
9. Toasted English : R. K. Narayan
10. The Portrait of a Lady : Khushwant Singh
11. Discovering Babasaheb : Ashok Mahadevan

Unit-II

Comprehension

Unit-III

Composition and Paragraph Writing (Based on expansion of an idea).

Unit-IV

Basic Language Skills : Vocabulary – Synonyms, Antonyms, Word Formation, Prefixes and Suffixes, Words likely to be confused and Misused, Words similar in Meaning or Form, Distinction between Similar Expressions, Speech Skills

Unit-V

Basic Language Skills: Grammar and usage – The Tense Forms, Propositions, Determiners and Countable/Uncountable Nouns, Verb, Articles, Adverbs.

Prescribed Books: English Language and Indian Culture, Published by M.P. Hindi Grant Academy.

Note :- Eight questions to be set from unit-1 and four to be attempted.



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B.C.A./B.A./ B.Com./B.Sc

Semester – I

Course	Subject	Subject Code
B.Sc. (PCM)	Development of Entrepreneurship m ferk fodk l	FC-101/2

Unit - I

Entrepreneurship- Definition, Characteristics and importance, Types and functions of an entrepreneur, merits of a good entrepreneur motivational factors of entrepreneurship.

उद्यमिता-परिभाषा, विशेषताएँ एवं महत्व, एक उद्यमी के प्रकार एवं कार्य, एक अच्छे उद्यमी के गुण, उद्यमिता अभिप्रेरणा
?KVdA

Unit - II

- Motivation to achieve targets and establishment of ideas. Setting targets and facing challenges. Resolving problems and creativity. Sequenced planning and guiding capacity, Development of self confidence.
 - Communication skills, Capacity to influence, leadership.
- v- y{; ikflr dh ij.kk ,o fopkjk dh LFkkiukA y{; fu/kkj.k ,o pukrh dk lkeukA
leL;k lek/kku ,o l'tukRedrka क्रमबद्ध योजना एवं क्षमता की दिशाबद्धता।
आत्मविश्वास का विकास।
- c- lEi:"k.k dykA iHkkfor dju: dh {kerkA urRo

Unit - III

- Project Report - Evaluation of selected process. Detailed project report – Preparation of main part of project report pointing out necessary and viability.
 - Selecting the form of Organization – Meaning and characteristics of sole Proprietorship, Partnership and cooperative committees, elements affecting selection of a form of an organisation.
 - Economic management – Role of banks and financial institutions banking, financial plans, working capital-evaluation and management, keeping of accounts.
- v- ifj;ktuk ifronu
puh gbi ifØ;k dk eY;kdu
विस्तृत परियोजना प्रतिवेदन- आवश्यकता एवं प्रासंगिकता परियोजना प्रपत्र के प्रमुख भाग परियोजना प्रतिवेदन
ri;kj djukA
- c- संगठन के प्रकार का चयन-एकाकी व्यवसाय, साझेदारी एवं सहकारी समिति का अर्थ एवं विशेषताएं संगठन के
p;u dk iHkkfor dju: oky: ?KVdA
- l- vkffkd ic/ku



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fofRr; I Lfku ,o cidk dh Hkfedk] cifdx] fofRr; ;ktuk] dk;dkjh iith&eY;kdu rFkk ic/ku] ykxr o eY; fu/kkjk rFkk ykHk dk eY;kdu] vkfFkd y[kk&tk[kk j[kukA

Unit - IV

- (a) Production management. Methods of purchase. Management of movable assets/goods. Quality management. Employee management. Packaging.
 - (b) Marketing management. Sales and the art of selling. Understanding the market and market policy. Consumer management. Time management.
- v- mRiknu dk ic/ku] [kjnhu d rjhd] py lEifRr@eky dk ic/ku] x.koRrk ic/ku] depkjh ic/ku] ifdx
- c- foi.ku ic/ku] fcØh ,o cpu dh dyk] cktkj dh le> ,o foi.ku uhfr] miHkkDrk ic/ku] le; ic/ku

Unit - V

- (a) Role of regulatory institutions – district industry centre, pollution control board, food and drug administration, special study of electricity development and municipal corporation.
 - (b) Role of development organizations, khadi & village Commission/ Board, MP Finance Corporation, scheduled banks, MP Women's Economics Development Corporation.
 - (c) Self-employment-oriented schemes, Prime Minister's Employment schemes, Golden Jubilee Urban environment scheme, Rani Durgavati Self-Employment scheme, Pt. Deendayal Self-employment scheme.
 - (d) Various grant schemes – Cost-of-Capital grant, interest grant, exemption from entry tax, project report, reimbursement grant, etc.
 - (e) Special incentives for women entrepreneurs, prospects & possibilities.
 - (f) Schemes of M.P. Tribal Finance Development Corporation, schemes of M.P. Antyavasai Corporation, schemes of M.P. Backward Class and Minorities Finance Development Corporation.
- 1- नियामक संस्थाओं की भूमिका—जिला उद्योग केन्द्र, प्रदूषण निवारण मंडल, खाद्य एवं औषधि प्रशासन, विद्युत विभाग तथा नगर निगम का विशेष अध्ययन।
 - 2- विकासात्मक संस्थाओं की भूमिका, खादी एवं ग्रामीण आयोग/बोर्ड, मध्य प्रदेश वित्त निगम, अनुसूचित बैंक, मध्य प्रदेश का महिला आर्थिक विकास निगम।
 - 3- Lojktxkj eyd ;ktuk, i & i/kkue=h jktxkj ;ktuk] Lo.k t;irh 'kgjh jktxkj ;ktuk] jkuh nxkorh Lojktxkj ;ktuk] fnun;ky Lojktxkj ;ktukA
 - 4- विभिन्न अनुदान योजनाएँ— लागत पूँजी अनुदान, ब्याज अनुदान, प्रवेश कर से छूट, परियोजना प्रतिवेदन, प्रतिपूर्ति vunku vkfnA
 - 5- महिला उद्यमियों हेतु विशेष प्रेरणाएँ, संभावनाएँ एवं समस्याएँ।
 - 6- मध्य प्रदेश आदिवासी िRr fodkl fuxe dh ;ktuk, i] e-i- vUR;kolk;h fuxe dh ;ktuk] e-i- fiNMk ox; ,o vYi i[];d foRr fodkl fuxe dh ;ktuk, iA



RKDF UNIVERSITY, BHOPAL

B.Sc.(Maths)

Semester – I

Branch	Subject title	Subject code
B.Sc (Mathematics)	Algebra and Trigonometry	BMM-111

UNIT-I

Linear Algebra:- Rank of a Matrix, Eigen values, Eigen vectors, Characteristics equation of a matrix, Cayley Hamilton theorem and its use in finding inverse of matrix, Application of matrix to a system of linear (both homogenous and non-homogenous) equations, theorems on consistency and inconsistency of a system of linear equations, Solving the linear equations with three unknowns.

UNIT-II

Theory of Equations . Relation between the roots and coefficients of a general polynomial equation in one variable, Transformation of equations, sum of r^{th} powers of roots; Reciprocal equations; Descartes rule of Sign.

UNIT-III

Expansion of Trigonometrical function- Expansions of $\sin x$, $\cos x$, $\tan x$ in terms of x ; $\sin nx$, $\cos nx$, $\tan nx$, $\sin^n x$, $\cos^n x$, $\tan^n x$, hyperbolic and inverse hyperbolic functions – simple problems.

UNIT-IV

Group Theory and number theory :- Equivalence relations; Groups; subgroups – cyclic groups and properties of cyclic groups – simple problems;

UNIT-V

Boolean Algebra and graph theory:- Algebra of logic, Boolean algebra, Basic theorem, Construction of truth table, Boolean function and Boolean expression, Switching circuit, graph theory and Application of graphs.

Reference Books :-

1. Algebra : T. K. Manickavachagam Pillal and others (S. Viswanathan publications)
2. Trigonometry : P. Duraipandian
3. Plane Trigonometry part 2 : S. L. Loney, (Macmillan and Co. London)



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B.Sc.(Maths)

Syllabus Semester – I

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Maths)	BASED ON INORGANIC, ORGANIC AND PHYSICAL CHEMISTRY	BCH-111

UNIT-I

Covalent Bond

Valence bond theory and its limitations, directional characteristics of covalent bond, various types of hybridization and shapes of simple inorganic molecules and ions (BeF₂, BF₃, CH₄, PF₅, SF₆, IF₇ SO₄, ClO₄)Valence shell electron pair repulsion (VSEPR)theory to NH₃, H₃O⁺, SF₄, ClF₃, ICl₂⁻ and H₂O. MO theory of hetero -nuclear (CO and NO) diatomic molecules, bond strength and bond energy.

UNIT-II

Gaseous States

Maxwell's distribution of velocities and energies (derivation excluded) Calculation of root mean square velocity, average velocity and most probable velocity. Collision diameter, collision number, collision frequency and mean free path. Deviation of Real gases from ideal behavior. Derivation of Vander Waal's Equation of State.

UNIT-III

Critical Phenomenon:

Critical temperature, Critical pressure, Critical volume and their determination. PV isotherms of real gases,.

Liquid States -Structure of liquids. Properties of liquids – surface tension, viscosity vapour pressure and optical rotations and their determination.

Solid State

Classification of solids, Laws of crystallography – (i) Law of constancy of interfacial angles (ii) Law of rationality of indices(iii) Law of symmetry. Definition of unit cell & space lattice. X ray diffraction by crystals. Derivation of Bragg equation. Liquid crystals :,types of liquid crystals. Applications of liquid crystals.

UNIT-IV

Stereochemistry of Organic Compounds-I

Concept of isomerism. Types of isomerism. elements of symmetry, molecular chirality,enantiomers, stereogenic centre, , chiral and achiral molecules with two stereogeniccentres, diastereomers, mesocompounds, resolution of enantiomers, inversion, retention and racemization.

Stereochemistry of Organic Compounds-II

Sequence rules, R & S systems of nomenclature. Geometric isomerism determination of configuration of geometric isomers. E & Z system of nomenclature, isomerism conformational



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analysis of ethane and n-butane, conformations of cyclohexane, Newman projection and Sawhorse formulae.

UNIT- V

Mechanism of Organic Reactions

Curved arrow notation, drawing electron movements with arrows, half-headed and double-headed arrows, homolytic and heterolytic bond breaking. Types of reagents – electrophiles and nucleophiles. Types of organic reactions. Energy considerations. Reactive intermediates carbocations, carbanions, free radicals, carbenes, arynes and nitrenes (formation, structure & stability). Assigning formal charges on intermediates and other ionic species.

Books:

1. Inorganic Chemistry, Huhee, Keiter and Keiter, Addition-Wesley, 1993.
2. Applied Mathematics for Physical Chemistry, J.R. Barante, Prentice Hall.
3. I. L. Levine, Quantum Chemistry, Prentice-Hall Inc., New Jersey.
4. T. Engel and P. Reid, Physical Chemistry, Benjamin-Cummings.
5. F. A. Carey and R. J Sundberg, Advanced Organic Chemistry, Part A and B, Plenum
6. P. S. Kalsi., Organic Reactions and their Mechanisms, New Age International

Paper (Practical's)

Section-A (Inorganic)

Volumetric Analysis

1. **Redox titrations:** Determination of Fe^{2+} , $\text{C}_2\text{O}_4^{2-}$ (using KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$)
2. **Iodometric titrations:** Determination of Cu^{2+} (using standard hyposolution).
3. **Complexometric titrations:** Determination of Mg^{2+} , Zn^{2+} by EDTA.

Section-B (Physical)

1. To determine the specific reaction rate of the hydrolysis of methyl acetate/ethyl acetate catalyzed by hydrogen ions at room temperature.
2. To prepare arsenious sulphide sol and compare the precipitating power of mono-, bi – and trivalent anions.

Section – C (Organic)

1. Preparation and purification through crystallization or distillation and ascertaining their purity through melting point or boiling point
2. (i) Iodoform from ethanol (or acetone)
(ii) *m*-Dinitrobenzene from nitrobenzene (use 1:2 conc. HNO_3 - H_2SO_4 mixture if fuming HNO_3 is not available)



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Syllabus Semester – I

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Maths)	MECHANICS	BPY 111

UNIT I

Projectile: Definition of Range, time of flight and angle of projection - Range up and down an inclined plane maximum range - two directions of projections for a given velocity and range. Impulse-Impact: Laws of impact – coefficient of restitution – impact of a smooth sphere on a fixed smooth plane – Direct impact between two smooth spheres – Loss of kinetic energy in direct impact – velocity change in oblique impact between two smooth spheres.

UNIT II

SHM: Composition of two SHM's of same period along a straight line and at the right angles to each other Lissajous figures. Dynamics of Rigid Bodies: Compound pendulum theory condition – for minimum period interchangeability of center of suspension and center of oscillation – g using compound pendulum - Bifilar pendulum – parallel and non – parallel threads.

UNIT III

Center of gravity: Center of gravity of a solid cone, Solid hemisphere, hollow hemisphere and a tetrahedron. Friction: Laws of friction - angle of friction – resultant reaction and cone of friction – equilibrium of a body on an inclined plane under the action of a force.

UNIT IV

Center of pressure: Definition – center of pressure of a rectangular lamina and triangular lamina. Hydrodynamics: equation of continuity of flow – Bernoullie's theorem – venturimeter – Pitot's tube. Conduction and Radiation : Thermal Conductivity - definition - thermal conductivity of a bad conductor - Lee's disc method - good conductor - Searle's method - radiation - Blackbody radiation – definition - Wien's Displacement law - Rayleigh Jean's law - Planck's law - Stepan's law and experimental verification of Stepan's law - Solar constant - temperature of the sun - by Angstrom's Pyrheliometer.

UNIT V

Maxwell's Thermodynamic relations : Derivation of Maxwell's thermodynamic relations - Helmholtz function - Gibb's function - Enthalpy – T-ds equation - Clausius - Clapeyron's Latent heat equation - specific heat relations.

Books for Study:

1. Mechanics and Mathematical Methods by R. Murugesan. S.Chand and Co.
2. Dynamics by M. Narayanamurthi and M. Nagarathnam, The National Publishing Company.
3. Statics, Hydrostatics and Hydrodynamics By Narayanamurthi and M. Nagarathnam, The National Publishing Company.



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SCHEME

Semester – II

No	Subject Code	Subject Type	Subject Title	Marks Allotted						
				Assignment Marks		Theory Marks		Practical Marks		Total Marks
				Max	Min	Max	Min	Max	Min	
1	FC-201/1	Core	हिन्दी भाषा संरचना	10	4	40	14	-	-	50
2	FC-201/2	Core	Environment	10	4	40	14	-	-	50
3	BMM 121	Core	Calculus and Numerical Analysis	20	8	80	27	-	-	100
4	BCH 121	Core	Based on Inorganic, Organic And Physical Chemistry	20	8	80	27	50	17	150
5	BPY 121	Core	Properties of Matter And Sound	20	8	80	27	50	17	150
Total				80	32	320	109	100	34	500



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B.C.A./B.A./ B.Com./B.Sc

Semester – II

Course	Subject	Subject Code
B.Sc. (Biotechnology)	हिन्दी भाषा संरचना	FC-201/1

इकाई –1

भारत वंदना (काव्य)	:	सूर्यकांत त्रिपाठी 'निराला'
जाग तुझको दूर जाना	:	सुश्री महादेवी वर्मा
स्वतंत्रता पुकारती (काव्य)	:	जयशंकर 'प्रसाद'
हम अनिकेतन (काव्य)	:	बालकृष्ण शर्मा 'नवीन'
		भाषा की महत्ता और उसके विविध रूप
		भाषा-कौशल

इकाई –2

करुणा (निबंध)	:	आचार्य रामचन्द्र शुक्ल
समन्वय की प्रक्रिया (निबंध)	:	रामधारी सिंह 'दिनकर'
बिच्छी बुआ (कहानी)	:	डॉ. लक्ष्मण विष्ट 'बटरोही'
अनुवाद	:	परिभाषा प्रकार, महत्व, विशेषताएं
		हिन्दी की शब्द-संपदा
		परिभाषिक शब्दावली

इकाई-3

विलायत पहुँच ही गया (आत्मकथांश)	:	महात्मा गांधी
अफसर (व्यंग्य)	:	शरद जोशी
तीर्थयात्रा (कहानी)	:	डॉ. मिथिलेश कुमारी मिश्र
मकड़ी का जाला (व्यंग्य)	:	डॉ. रामप्रकाश सक्सेना
		वाक्य-संरचना: तत्सम, तद्भव देशज विदेशी

इकाई-4

अप्य दीपो भव (वक्तृत्व कला)	:	स्वामी श्रद्धानन्द
भारत का सामासिक व्यक्तित्व (प्रस्तावना)	:	जवाहरलाल नेहरू
पत्र मैसूर के महाराजा को (पत्र-लेखन)	:	स्वामी विवेकानन्द
बनी रहेगी किताबें (आलेख)	:	डॉ. सुनीता रानी घोष
		पत्र-लेखन : महत्व और उसके विविध रूप
		सड़क पर दौड़ते ईहा मृग (निबंध) डॉ. श्यामसुन्दर दुबे

इकाई-5

योग की शक्ति (डायरी)	:	डॉ. हरिवंशराय बच्चन
कोश के अखाड़े में कोई पहलवान नहीं उतरता (साक्षात्कार)	:	भाषाविद् डॉ. हरदेव बाहरी से प्रो. त्रिभुवननाथ शुक्ल
नीग्रो सैनिक से भेंट (यात्रा-संस्मरण)	:	डॉ. देवेन्द्र सत्यार्थी
यदि बा न होती तो शायद गांधी को यह ऊँचाई न मिलती (साक्षात्कार)	:	5 कथाकार गिरिराज किशोर से सत्येन्द्र शर्मा
सर-लेखन, भाव-पल्लवन साक्षात्कार प्रयोजन और कौशल निर्धारित पाठ्य पुस्तक	:	"हिन्दी भाषा संरचना" मध्यप्रदेश हिन्दी ग्रन्थ अकादमी द्वारा प्रकाशित



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Semester – II

Course	Subject	Subject Code
B.Sc. (Biotechnology)	Environment @ i ;koj .k	FC-201/2

Unit-I

Study of Environmental and ecology:

- (a) Definition and Importance. (b) Environmental Pollution and problems.
 (c) Public participation and Public awareness.
 पर्यावरण एवं पारिस्थितिकीय अध्ययन
 (क) परिभाषा एवं महत्व (ख) पर्यावरण प्रदूषण एवं समस्याएँ
 (ग) जनभागीदारी एवं जन जागरण

Unit-II

Environmental Pollution :

- (a) Air, water, noise, heat and nuclear pollution.
 (b) Causes, effect and prevention of pollution.
 (c) Disaster management – Flood, Earthquake, cyclones and landslides.
 पर्यावरणीय प्रदूषण
 (क) वायु, जल, ध्वनि, ताप एवं आणविक-प्रदूषण (ख) प्रदूषण के कारण, प्रभाव एवं रोकथाम
 (ग) आपदा प्रबंधन- बाढ़, भूकंप, चक्रवात एवं भूस्खलन

Unit-III

Environment and social problems :

- (a) Development – non-sustainable to Sustainable. (b) Energy problems of cities.
 (c) Water preservation – rain-water collection.
 पर्यावरण और सामाजिक समस्याएँ
 (क) अधारणीय से धारणीय विकास (ख) नगरों की ऊर्जा समस्या
 (ग) जल संरक्षण- वर्षा, जल-संग्रहण

Unit-IV

Role of mankind in conserving natural resources :

- (a) Food resources – World food problem (b) Energy resources – increasing demand for energy.
 (c) Land resources – Land as resources.
 प्राकृतिक संसाधनों के संरक्षण में मनुष्य की भूमिका
 (क) खाद्य-आहार संसाधन – विश्व आहार समस्या (ख) ऊर्जा संसाधन- ऊर्जा की बढ़ती मांग
 (ग) भूमि संसाधन- भूमि संसाधन के गुप में, भूमि अवनयन, मनुष्यकृत भूस्खलन

Unit-V

Environment conservation laws :

- (a) Conservation laws for air and water pollution.
 (b) Wildlife conservation laws.
 (c) Role of information technology in protecting environment & health.
 पर्यावरण संरक्षण कानून
 (क) वायु तथा जल प्रदूषण-संरक्षण कानून
 (ख) वन्य प्राणि संरक्षण कानून
 (ग) पर्यावरण तथा स्वास्थ्य रक्षा में सूचना प्रौद्योगिकी की भूमिका



RKDF UNIVERSITY, BHOPAL

B.Sc.(Maths)

Semester – II

Branch	Subject title	Subject code
B.Sc (Mathematics)	Calculus and Numerical Analysis	BMM-121

Unit-1

Successive differentiation, Leibnitz theorem, Maclaurin and Taylor series expansions, Curvature, Tests for concavity and convexity, Points of inflexion, Multiplepoints, Tracing of curves in Cartesian co-ordinates.

Unit-2

Limit and continuity of functions of two variables, Introduction of Partial differentiation, Euler's Theorem on homogeneous function, Jacobians, Differentiability of real-valued functions of two variables, Taylor's theorem for functions of two variables, Double and triple integrals, Dirichlet's integrals.

Unit-3

Numerical integration: General quadrature formula, Trapezoidal rule, Simpson's one-third rule, Simpson's three-eighth rule, Weddle's rule, Euler-Maclaurin Summation formula, Sterling's formula for $n!$.

Unit-4

Linear Differential equations and equations reducible to the linear form, Exact differential equation, First order and higher degree equations Solvable for x , y and p , Clairaut's form and singular solutions, linear differential equations with constant coefficients.

Unit-5

Vector differentiation, Gradient, Divergence and Curl, Vector integration, Theorem of Gauss (without proof) and problems based on it, Theorem of Green (without proof) and problems based on it, Stoke's theorem (without proof) and problems based on it.

Texts Book

1. Gorakh Prasad – Differential Calculus, Pothishala pvt. Ltd. Allahabad
2. Gorakh Prasad – Integral Calculus, Pothishala pvt. Ltd. Allahabad
3. D.A. Murray : Introductory Course in Differential Equations, Orient Long man, India 1967.
4. N. Saran & S.N. Nigam – Introduction to Vector Analyss, Pothishala Pvt. Ltd., Allahabad.

Reference Book

1. P.K. Jain and S. K. Kaushik, An introduction of Real Analysis, S.Chand & Co. New Delhi 2000.
2. Calculus of finite differences and Numerical Analysis by Gupta-Malik, Krishna prakastan Mandir, Meerut.
3. Numerical Analysis by B.D. Gupta, Konark publishing.



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Syllabus Semester – II

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Mathematics)	Based on inorganic, organic and physical chemistry	BCH-121

UNIT-I

Hydrogen Bonding – Definition, Types, effects of hydrogen bonding on properties of substances, application Brief discussion of various types of Vander Waals Forces

Metallic Bond and Semiconductors

Metallic Bond- Brief introduction to metallic bond, band theory of metallic bond
Semiconductors- Introduction, types and applications.

UNIT-II

Rate of reaction, rate equation, factors influencing the rate of a reaction – concentration, temperature, pressure, solvent, light, catalyst. Order of a reaction, integrated rate expression for zero order, first order, second and third order reaction. Half life period of a reaction. Methods of determination of order of reaction.

Kinetics-II

Effect of temperature on the rate of reaction – Arrhenius equation. Theories of reaction rate – Simple collision theory for unimolecular and bimolecular collision. Transition state theory of Bimolecular reactions.

UNIT-III

Electrolytic conduction, factors affecting electrolytic, Arrhenius theory of ionization, Ostwald's Dilution Law. Debye-Huckel – Onsager's equation for strong electrolytes (elementary treatment only) Transport number, definition and determination by Hittorf's methods, (numerical included). **Electrochemistry-II** Kohlrausch's Law, calculation of molar ionic conductance and effect of viscosity. Application of Kohlrausch's Law in calculation of conductance of weak electrolytes at infinite dilution. Applications of conductivity, Definition of pH and pKa, Buffer solution, Buffer action, Henderson – Hazelequation, Buffer mechanism of buffer action.

UNIT-IV

Nomenclature of alkenes, mechanisms of dehydration of alcohols and dehydrohalogenation of alkyl halides. The Saytzeff rule, Hofmann elimination, physical properties and relative stabilities of alkenes. Chemical reactions of alkenes mechanisms involved in hydrogenation, electrophilic and free radical additions, Markownikoff's rule, hydroboration-oxidation, oxymercuration reduction, ozonolysis, hydration, hydroxylation and oxidation with KMnO_4 .

UNIT-V

IUPAC nomenclature of branched and unbranched alkanes, the alkyl group, classification of carbon atoms in alkanes. Isomerism in alkanes, sources, methods of formation (with special reference to Wurtz reaction, Kolbe reaction Cycloalkanes nomenclature, synthesis of



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cycloalkanes and their derivatives –photochemical (2+2) cycloaddition reactions, dehalogenation of β,β -dihalides, ,pyrolysis of calcium or barium salts of dicarboxylic acids, Baeyer's strain theory and its limitations.

Book suggested :

1. Advanced Inorganic Chemistry, F.A. Cotton and Wilkinson, John Wiley.
2. Chemistry of the Elements. N.N. Greenwood and A. Earnshaw, Pergamon.
3. J. March., Advanced Organic Chemistry: Reactions, Mechanisms and Structure, John Wiley.
4. P. S. Kalsi., Organic Reactions and their Mechanisms, New Age International
5. J. P. Lowe and K.Peterson, Quantum Chemistry Academic Press.

Paper (Practical's)

Section-A (Inorganic)

Paper Chromatography

Qualitative Analysis of the any one of the following Inorganic cations and anions by paper chromatography (Pb^{2+} , Cu^{2+} , Ca^{2+} , Ni^{2+} , Cl^- , Br^- , I^- and PO_4^{3-} and NO_3^-

Section-B (Physical)

1. To determine the surface tension of a given liquid by drop number method.
2. To determine the viscosity of a given liquid.
3. To determine the specific refractivity of a given liquid.

SECTION – C (Organic)

1. Dibenzalacetone from acetone and benzaldehyde
2. Aspirin from salicylic acid. To study the process of) sublimation of camphor and phthalic acid

Books for practical:

1. Vogel's Textbook of Quantitative Analysis, revised, J. Bassett, R.C. Denney, G.H. Jeffery and J. Mendham, ELBS.
2. Synthesis and Characterization of Inorganic Compounds, W.L. Jolly. Prentice Hall.
3. Experiments and Techniques in Organic Chemistry, D.P. Pasto, C. Johnson and M.Miller, Prentice Hall.
4. Macroscale and Microscale Organic Experiments, K.L. Williamson, D.C. Heal



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B.Sc.(Maths)

Syllabus Semester – II

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Mathematics)	PROPERTIES OF MATTER AND SOUND	BPY 121

UNIT-I

Elasticity: Three types of elastic moduli - Poisson's ratio - Bending of beams - Expression for bending moment - Depression of the loaded end of a Cantilever - uniform - non uniform bending - theory - experiment pin and microscope method - work done in uniform bending – Koenig's method – non-uniform bending - theory - expression for couple per unit twist - determination of rigidity modulus - Static torsion method with scale and telescope - Rigidity modulus by torsion pendulum with mass.

UNIT-II

Interference and Interferometers : Coherence - temporal coherence and spatial coherence - Air wedge – testing the planeness of a surface - Michelson Interferometer - types of fringes - Difference in wavelength of Sodium D1, D2 lines and thickness of a thin transparent plate. Multiple beam interference - Feby - Perot interferometer - formation of fringes. Holography : Holography - recording and reconstruction.

UNIT-III

Surface tension and Osmosis : Surface energy - angle of contact and its determination - excess of pressure inside curved surface - formation of drops - Experimental study of variation of Surface tension with temperature - drop weight method of determining surface tension and interfacial surface tension - angle of contact of mercury - Quincke's method - surface tension and vapour pressure osmosis - experimental determination of osmotic pressure - Laws of osmosis pressure - osmotic and vapour pressure of a solution.

UNIT-IV

Sound: Definition of free, damped and forced vibrations – Theory of forced vibrations - Resonance - Sharpness of resonance - Fourier's theorem - application for Saw- tooth wave and square wave. - Sonometer - determination of A.C. frequency using sonometer - Determination of frequency using Melde's apparatus.

UNIT-V

Ultrasonics: Ultrasonics - Production - Piezo electric method – magneto-striction method - detection - properties - applications. Acoustics : Acoustics of buildings - reverberation time - derivation of Sabine's formula - determination of absorption coefficient.

BOOKS FOR STUDY:

1. Elements of properties of matter by D. S. Mathur S. Chand & Co., (2005).
2. Properties of matter by R. Murugesan, S. Chand & Co., (2005).



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3. Properties of matter by Brijlal and N. Subramaniam S. Chand & Co., (2005).
4. Properties of matter and Acoustics by R. Murugesan, S. Chand & Co., (2005).
5. A Text Book of Sound by N. Subramaniam and Brijlal, S. Chand & Co., (2005).

BOOKS FOR REFERENCE:

1. Fundamentals of General Properties of Matter, H. R. Gulati, S. Chand & Co., (2005).
2. Properties of Matter, Subramania Iyer and Ranga Rajan, Viswanathan Publication (2002).
3. A Text Book of Sound (2005), R. L. Saighal, S. Chand & Co.,
3. Geometrical and Physical Optics - P. K. Chakrabarti, New Central Book Agency (P) Ltd, Kolkata., 2005.
4. Optics - D.R. Khanna and H.R. Gulati, R. Chand & Co, New Delhi., 1979.
5. Engineering Physics - G. Vijayakumari, Vikas Publications.
- : 1. Optics - Eugene Hecht, Fourth Edition, Pearson Education, New Delhi. 2007.
2. Fundamentals of Optics - Jerkins A Francis and White E Harvey, McGraw Hill Inc., New Delhi, 1976.
3. Optical Physics - S.G. Lipson, H. Lipson and D.S. Tannhauser, Cambridge University Press. 1995.
4. Fundamentals of Optics - M.G. Raj, Anmol Publications Pvt. Ltd., New Delhi, 1996.

SEMESTER-II

PRACTICALS PRACTICAL

01. Young's modulus (q) - uniform bending - pin and microscope.
02. Young's modulus (q) - uniform bending - scale and telescope method.
03. Static Torsion – Rigidity modulus.
04. Torsion Pendulum – Moment of Inertia and Rigidity modulus – symmetrical masses.
05. Coefficient of Viscosity of a liquid - graduated burette - radius by mercury pellet method.
06. Melde's apparatus - frequency - transverse and longitudinal modes.
07. Specific heat capacity of a liquid by cooling - verification of Newton's law of cooling.
08. Air Wedge - thickness of a wire and its insulation.
09. Spectrometer - grating - minimum deviation – Determination of wavelength of mercury lamp.
10. Potentiometer - ammeter calibration.
11. Potentiometer - Specific resistance of the given coil and length of second coil without unwinding.
12. M and BH - Deflection Magnetometer - $TAN A$ and $TAN B$ position.
13. Field along the axis of a coil - deflection magnetometer – determination of BH .
14. Carey-Foster's bridge - Specific resistance of a coil.
15. BG - Comparison of Capacities.
16. BG - Comparison of EMF's of two cells.
17. Zener diode – Voltage regulator using four diodes and percentage of regulation.
18. Verification of De Morgan's theorem.



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SCHEME

Semester – III

No	Subject Code	Subject Type	Subject Title	Marks Allotted						
				Assignment Marks		Theory Marks		Practical Marks		Total Marks
				Max	Min	Max	Min	Max	Min	
1	FC-301/1	Core	Values & Spirituality	10	4	40	14	-	-	50
2	FC-301/2	Core	Environment	10	4	40	14	-	-	50
3	BMM 131	Core	Linear Algebra	20	8	80	27	-	-	100
4	BCH 131	Core	Based on Inorganic, Organic And Physical Chemistry	20	8	80	27	50	17	150
5	BPY 131	Core	Electricity and Magnetism	20	8	80	27	50	17	150
Total				80	32	320	109	100	34	500



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Semester – III

Course	Subject	Subject Code
B.Sc (Mathematics)	Values & Spirituality	FC-301/1

Chapter 1: VALUE EDUCATION

1.1	Objectives	1.2	Introduction	1.3	Concepts of Values
1.4	Definition and Types of values			1.5	The need for Education in values
1.6	Challenges for Value adoption			1.7	Character development
1.8	Vision of a better world			1.9	Summary
1.10	Glossary			1. 11	Suggested reading

Chapter 2: INCULCATION OF VALUES

2.1	Objectives	2.2	Introduction	2.3	Classification of values
2.4	Personal Values	2.5	Family Values	2.6	Social Values
2.7	Spiritual values	2.8	Benefits of value adoption	2.9	Summary
2.10	Glossary	2.11	Suggested reading		

Chapter 3: MAJOR RELIGIONS OF THE WORLD

3.1	Objectives	3.2	Introduction	3.3	Hinduism
3.4	Jainism	3.5	Buddhism	3.6	Christianity
3.7	Islam	3.8	Sikhism	3.9	Summary
3.10	Glossary	3.11	Suggested reading		

Chapter 4: EXPLORING THE SELF

4.1	Objectives	4.2	Introduction	4.3	Anatomy of the self
4.4	The cyclic processes within the self			4.5	States of the awareness
4.6	Innate qualities	4.7	Acquired qualities	4.8	Empowering the self
4.9	Summary	4.10	Glossary	4.11	Suggested reading

Chapter 5: THOUGHT AND THE THINKER

5.1	Objectives	5.2	Introduction	5.3	Know the mind(Team)
5.4	Thought power	5.5	Types of thoughts	5.6	Thinking process
5.7	Positive thinking	5.8	Power and Acts	5.9	Summary
5.10	Glossary	5.11	Suggested reading		



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Semester – II

Course	Subject	Subject Code
B.Sc (Mathematics)	Environment	FC-301/2

Unit - I

Problems of natural resources:

- Problems of water resources- Utilization of surface and ground water overutilization, flood, draught, conflict over water, Dams and related problems.
- Problems of forest resources -uses and over utilization, deforestation, Dams and its effects on forests and tribes.
- Problems of land resources -land as a source , erosion of land, man-induced, land slides and desertification.

प्राकृतिक संसाधन की समस्याएँ

- जल संसाधन की समस्या— सतह एवं भूजल का उपयोग , अति दोहन बाढ़ सूखा , जल पर संघर्ष , बाँध —लाभ एवं समस्याएँ
- वन संसाधन की संस्याएँ —उपयोग एवं अति दोहन , वनोन्मूलन , इमारती लकड़ी, निस्सारण , बाँध एवं उनका बन और आदिवासीयो पर प्रभाव
- भूमि संसाधन की समस्याएँ— स्त्रोत के क्रय में भूमि का अवभ्रमण , मानव प्रेरित भू — स्खलन और मरुस्थीकरण

Unit - II

Bio-diversity and its protection -

- Value of bio-diversity -consumable use: productive use, social, alternative, moral asthetic and values
- Bio - diversity and multi -diversity at global and national levels.
- Threats to bio -diversity -loss of habitat, poaching of wildlife, man wildlife conflicts.

जैव विविधता और उसका संरक्षण –

- जैव विविधता का मूल्य – उपभोग्य , उपयोग , उत्पादक उपयोग, सामाजिक, नैतिक सौर्दयंगत तथा वैकल्पिक मूल्य
- वैश्विक , राष्ट्रीय तथा स्थानीय स्तरो पर जैव विविधता वृहत विविधताओ के राष्ट्र रूप में भारत ।
- जैव विविधता के खतरे – आवासीय हानि ,वन्य जीवन में अनाधिकार घुसपैठ तथा मानव , वन्य जीवन —संघर्ष

Unit - III

Human population and environment

- Population growth, disparities between countries.
- Population explosion, family welfare programme
- Environment and human health

पर्यावरण और जनसंख्या तथा पर्यावरण

- जनसंख्या – वृद्धि , राष्ट्रों के बीच अंतर
- जनसंख्या —विस्फोट , परिवार कल्याण कार्यक्रम
- पर्यावरण और मानव स्वास्थ्य



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B.Sc.(Maths)

Unit - IV

Multidisciplinary nature of environmental studies :

- (a) Natural resources
- (b) Social problems and the environment
- (c) Environmental awareness

पर्यावरण और उसका बहुअनुशासनिक स्वरूप

- (क) प्राकृतिक संसाधन
- (ख) सामाजिक समस्याएँ और पर्यावरण
- (ग) पारिस्थिति तन्त्र

Unit - V

Multidisciplinary nature of environmental studies :

- (a) Natural resources
- (b) Social problems and the environment
- (c) Environmental awareness

पर्यावरण और उसका बहुअनुशासनिक स्वरूप

- (क) प्राकृतिक संसाधन
- (ख) सामाजिक समस्याएँ और पर्यावरण
- (ग) पारिस्थिति तन्त्र



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B.Sc.(Maths)

Semester – III

Branch	Subject Title	Subject code
B.Sc (Mathematics)	Linear Algebra	BMM-131

UNIT:-I

ALGEBRA : Partial fractions, binomial, exponential and logarithmic series (without proof) summation and approximation-simple problems.

UNIT:-II

THEORY OF EQUATIONS : Polynomial equations with real coefficients, irrational roots, complex roots, symmetric functions of roots, transformation of equation by increasing or decreasing roots by a constant, reciprocal equations. Newton's method to find a root approximately - simple problems.

FINITE DIFFERENCES : Operators E, difference tables, Newton's forward and backward interpolation formulae, Lagrange's interpolation formulae.

UNIT:-III

Laplace transformations: Laplace transformations, Linearity of the Laplace transformation, Existence theorem of Laplace transforms, Laplace transforms of derivatives and integrals, Shifting theorem, Differentiation and integration of transforms, Inverse Laplace transforms, Convolution theorem, Applications of Laplace, transformation in solving linear differential equations with constant coefficients

UNIT:-IV

DIFFERENTIAL CALCULUS: Series Solution of Differential Equations-Power series Method, Bessel's Equation, Bessel's function and its properties, recurrence and generating relations, Legendre's Equation, Legendre's function and its properties,

UNIT:-V

Groups :- Normal sub group, Quotient groups, homomorphism and isomorphism of groups, Kernel of homomorphism of groups, fundamental theorem of homomorphism of groups, Permutation groups (even and odd permutations), Alternating groups A_n , Cayley's theorem.

Reference Books:

- , S.K. Jain and S.R. Nagpaul, Basic Abstract Algebra, Wiley Eastern, New Delhi, 1997. 1.
T.M. Apostol Mathematical Analysis Narosa Publishing House New Delhi 1985.
2. Murray R. Spiegel, Theory and Problems of Advanced Calculus, Schaum Publishing Co. New York.



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3. N. Piskunov, Differential and Integral Calculus, Peace Publishers, Moscow.
4. S.C. Malik, Mathematical Analysis, Wiley Eastern Ltd. New Delhi.
5. P.B. Bhattacharya
6. I. S. Luther and I.B. S. Passi, Algebra Vol- I , II, Narosa Publishing House
7. R.R. Goldberg, Real Analysis, I.B.H. Publishing Co. New Delhi, 1970.



RKDF UNIVERSITY, BHOPAL

B.Sc.(Maths)

Syllabus Semester – III

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Mathematics)	Based on inorganic, and Physical Chemistry	BCH-131

UNIT-I

Definition of transition elements, position in the periodic table, General characteristics & properties of 1st transition elements, Structures & properties of some compounds of transition elements– TiO_2 , VOCl_2 , FeCl_3 , CuCl_2 and $\text{Ni}(\text{CO})_4$. **Chemistry of Elements of IInd & IIIrd transition series** General characteristics and properties of the IInd and IIIrd transition elements Comparison of properties of 3d elements with 4d & 5d elements with reference only to ionic radii, oxidation state, magnetic and Spectral properties and stereochemistry.

UNIT-II

Werner's coordination theory, effective atomic number concept, chelates, nomenclature of coordination compounds, isomerism in coordination compounds, valence bond theory of transition metal complexes. **Non-aqueous Solvents** Physical properties of a solvent, types of solvents and their general characteristics, reactions in non-aqueous solvents with reference to liquid NH_3 and liquid SO_2 .

UNIT-III

Definition of thermodynamic terms: system, surrounding etc. First law of thermodynamics: statement, definition of internal energy and enthalpy. Heat capacity, heat capacities at constant volume and pressure and their relationship. Joule's law – Joule – Thomson coefficient for ideal gas and real gas: and inversion temperature. **Thermodynamics-II** Calculation of w.q. dU & dH for the expansion of ideal gases under isothermal and adiabatic conditions for reversible process, Temperature dependence of enthalpy, Kirchhoff's equation. Bond energies and applications of bond energies.

UNIT-IV

Monohydric alcohols nomenclature, methods of formation by reduction of aldehydes, ketones, carboxylic acids and esters. Hydrogen bonding. Acidic nature. Reactions of alcohols. **Epoxides** Synthesis of epoxides. Acid and base-catalyzed ring opening of epoxides, orientation of epoxide ring opening.

UNIT-V

Nomenclature of Carboxylic acids, structure and bonding, physical properties, acidity of carboxylic acids, Preparation of carboxylic acids. Reactions of carboxylic acids. **Acid Derivatives** Structure, nomenclature and preparation of acid chlorides, esters, amides and acid anhydrides. Relative stability of acyl derivatives. Physical properties, inter conversion of acid derivatives by nucleophilic acyl substitution. Mechanisms of esterification and hydrolysis (acidic and basic).



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Book suggested :

1. R. G. Mortimer, Mathematics for Physical Chemistry Elsevier.
2. F. L. Pilar, Elementary Quantum Chemistry , Dover Publication
3. E. L. Eliel, Stereochemistry of Carbon Compounds, McGraw-Hill
4. S. M. Mukherji and S. P. Singh, Reaction Mechanism in Organic Chemistry,
5. Comprehensive Coordination Chemistry eds., G. Wilkinson, R.D. Gillars and J.A. McCleverty, Pergamon.

Paper (Practicals)

Section – I (Inorganic)

1. Gravimetric Analysis

Quantitative estimations of, Cu^{2+} as copper thiocyanate and Ni^{2+} as Ni – dimethylglyoxime.

Section-B (Physical)

1. To determine the CST of phenol – water system.
2. To determine the solubility of benzoic acid at various temperatures and to determine the ΔH of the dissolution process
3. To determine the enthalpy of neutralisation of a weak acid/weak base vs. strong base/strong acid and determine the enthalpy of ionisation of the weak acid/weak base.

Section-C (Organic)

Systematic identification (detection of extra elements, functional groups, determination of melting point or boiling point and preparation of at least one pure solid derivative) of the following simple mono and bifunctional organic compounds: Naphthalene , anthracene, acenaphthene, benzyl chloride, *p*-dichlorobenzene, *m*-dinitrobenzene, *p*-nitrotoluene, resorcinol , hydroquinone, 1-naphthol, 2-naphthol, benzophenone, ethyl methyl ketone, benzaldehyde, vanillin, oxalic acid, succinic acid,



RKDF UNIVERSITY, BHOPAL

B.Sc.(Maths)

Syllabus

Semester – III

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Mathematics)	ELECTRICITY AND MAGNETISM	BPY 131

UNIT- I

Principle of a capacitor - energy stored in a capacitor - energy density - change in energy due to dielectric slab - force of attraction between plates of a charged capacitor - capacitance of a spherical and cylindrical capacitors - types of capacitors - electrometers - Kelvin's attracted disc electrometer - quadrant electrometer - measurement of potential, ionization current and dielectric constant (solid).

UNIT - II

Positive ray analysis - Thomson's parabola method - theory - determination of e/m and mass of positive ions - Aston's mass spectrograph determination of masses of isotopes - uses of mass spectrographs – separation of isotopes - mass spectrograph method - diffusion method - thermal diffusion method - pressure diffusion method.

UNIT - III

Theory of alpha scattering - Rutherford scattering formula – experimental verification - nature of privileged quantum orbits - Bohr's correspondence principle - effect of motion of nucleus - evidences in favour of Bohr's theory - Determination of critical potential - Davis and Goucher's method - Sommerfield's relativistic atom model -application to fine structure of $H\alpha$ line.

UNIT -IV

Transient current - growth and decay of current in a circuit containing resistance and inductance - growth and decay of charge in a circuit containing resistance and capacitance - measurement of high resistance by leakage - growth and decay of charge in a LCR circuit - condition for the discharge to be oscillatory - frequency of oscillation - Importance in wireless telegraphy.

UNIT- V

Alternating current - peak, average and RMS value of current and voltage - form factor - j operator - ac circuit containing resistance and inductance - choke coil - ac circuit containing resistance and capacitance - series and parallel resonance circuits - Q factor - power in an ac circuit containing LCR – Wattless current - Transformer - construction, theory and uses - energy loss - skin effect – Tesla coil.

Books for Study:

1. Electricity and Magnetism - Brijlal and Subramaniam, S. Chand & Co., (2005).
2. Electricity and Magnetism - R. Murugesan, S. Chand & Co., (2005).

Books for Reference:

1. Electricity and Magnetism - D. N. Vasudeva, S. Chand & Co., (2005).
2. Electricity and Magnetism- K. K. Tewari, S. Chand & Co., (2005)



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SCHEME

Semester – IV

No	Subject Code	Subject Type	Subject Title	Marks Allotted						
				Assignment Marks		Theory Marks		Practical Marks		Total Marks
				Max	Min	Max	Min	Max	Min	
1	FC-401/1	Core	English Language	10	4	40	14	-	-	50
2	FC-401/2	Core	Development of Entrepreneurship	10	4	40	14	-	-	50
3	BMM 141	Core	Analysis	20	8	80	27	-	-	100
4	BCH 141	Core	Based on Inorganic, Organic And Physical Chemistry	20	8	80	27	50	17	150
5	BPY 141	Core	Electricity and Magnetismbasic Electronics	20	8	80	27	50	17	150
Total				80	32	320	109	100	34	500



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Semester – IV

Course	Subject	Subject Code
B.Sc (Mathematics)	English Language	FC-401/1

Unit-I

1. William Wordsworth : “The World is Too Much With Us”
2. K. Aludiapillai : “Communication Education and Information Technology”
“Democratic Decentralisation”
3. S. C. Dubey : “Basic Quality of Life”
4. Sister Nivedita : “The Judgment Seat of Vikramaditya”
5. Juliun Huxley : “War as a Biological Phenomenon”
6. Robert Frost : “Stopping by Woods on a Snowy Evening”
7. Ruskin Bond : “The Cherry Tree”

Unit-II

Short Essay of about 250-300 words.

Unit-III

Translation of a short passage from Hindi to English.

Unit-IV

Drafting CV, writing e-mail message for official purpose.

Unit-V

Language Skills, One-word substitution, homonyms, homophones, words that confuse, Punctuation, Idioms



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B.Sc.(Maths)

B.C.A./B.A./B.Com./B.Sc

Semester – IV

Course	Subject	Subject Code
B.Sc (Mathematics)	Development of Entrepreneurship	FC-401/2

Unit-I

Entrepreneurship - Meaning, Concept, Characteristics of entrepreneur.

उद्यमिता – का आशय, मत, उद्यमिता के गुण।

Unit-II

Types of entrepreneurship, importance and views of various thinkers (Scholars).

- Formation of goals, How to achieve goals.
- Problems in achieving targets and solution.
- Self motivation, elements of self motivation and development.
- Views of various scholars, evaluation, solutions.

Leadership capacity : Its development and results.

उद्यमिता के प्रकार, महत्व और विभिन्न विद्वानों के मत, लक्ष्य निर्माण, लक्ष्य कैसे प्राप्त करें। लक्ष्य प्राप्ति में समस्याएँ, उनका समाधान, स्वप्रेरणा, स्वप्रेरणा के तत्व और विकास, विभिन्न विद्वानों के मत, आकलन, निष्कर्ष, नेतृत्व समता, उसका विकास और प्रतिफलन

Unit-III

Projects and various organisations (Govt., non-Govt.), Govt. Projects, Non-Govt. projects. Contribution of Books, their limitations, scope.

परियोजनाएँ तथा विभिन्न संगठन (शासकीय-अशासकीय), शासकीय परियोजनाएँ- अशासकीय परियोजनाएँ- बैंको का योगदान, उनकी सीमाएँ, क्षेत्र

Unit-IV

Functions, qualities, management of a good entrepreneur. Qualities of the entrepreneur (Modern and traditional). Management skills of the entrepreneur. Motive factors of the entrepreneur.

अच्छे उद्यमी के कौन-कौनसे कार्य, गुण, प्रबंधन इत्यादि, अच्छे उद्यमी के गुण आधुनिक और पूर्ववर्ती, उद्यमी की प्रबंधन कला, उद्यमी के प्रेरक तत्व

Unit-V

Problems and Scope of the Entrepreneur :

- Problem of Capital
- Problem of Power
- Problem of registration
- Administrative problems
- Problems of Ownership.

उद्यमी की समस्याएँ, क्षेत्र, पूँजी की समस्या, शक्तिकरण की समस्या, पूँजीवन की समस्या, प्रशासनिक समस्याएँ, स्वामित्व की समस्याएँ, इत्यादि



RKDF UNIVERSITY, BHOPAL

B.Sc.(Maths)

Semester – IV

Branch	Subject title	Subject code
B.Sc (Mathematics)	Analysis	BMM-141

Unit-1

Group automorphisms, inner automorphism, Group of automorphisms, Conjugacy relation and centraliser, Normaliser, Counting principle and the class equation of a finite group, Cauchy's theorem for finite abelian groups and non-abelian groups.

Unit-2

Introduction to rings, subrings, integral domains and fields, simple properties and examples, ring homomorphism, ideals and quotient rings. Divided differences and Newton's divided differences formula for interpolation and Lagrange's formula for interpolation.

Unit-3

Laplace Transformation (LT) - definitions, LT of the function t , e^{at} , $\cos at$, $\sin at$, $e^{at} \cos bt$, $e^{at} \sin bt$, Transform $f'(t)$, $f''(t)$ - inverse LT relating to the above standard functions. Convolution theorem based problem
Maxima, Minima and saddle points of functions of two variables, Improper integrals and their convergence, Comparison test, Abel's and Dirichlet's tests,

Unit-4

Partial Differential equations of the first order, Lagrange's solution, Some special types of equations which can be solved easily by methods other than general methods, Charpit's general method of solution, Partial differential equations of second and higher orders, Partial differential equations reducible to equations with constant coefficients.

Unit-5

Continuity and differentiability of Complex functions, Analytical function, Cauchy Riemann equation, Harmonic function, Mobius transformations, fixed points, cross ratio.

Text Books :

1. I.N. Sneddon, Elements of partial Differential equations Mc Graw Hill, Co. 1988
2. Shanti Narayan, Theory of Functions of a Complex Variable, S. Chand & Co., New Delhi.
3. Calculus of finite differences and Numerical analysis by Gupta-Malik, Krishna



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Prakastan Mandir, Meerut.

4. Numerical Analysis by B.D. Gupta,

Reference Books:

1. T.M. Apostol, Mathematical Analysis Narosa Publishing House, New Delhi 1985
2. Shanti Narayan, A Text Book of Modern Abstract Algebra, S. Chand & Co. New Delhi
3. . P.B. Bhattacharya, S.K. Jain and S.R. Nagpaul, Basic Abstract Algebra, Wiley Eastern, New Delhi, 1997.
4. . I. S. Luther and I.B. S. Passi, Algebra Vol- I , II, Narosa Publishing House.
5. . R. V. Churchill & J.W. Brown, Complex Variables and Applications, 5th Edition,



RKDF UNIVERSITY, BHOPAL

B.Sc.(Maths)

Syllabus Semester – IV

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Mathematics)	BASED ON INORGANIC, AND PHYSICAL CHEMISTRY	BCH-141

UNIT-I

Electronic structure, oxidation states and ionic radii and lanthanide contraction, complex formation, occurrence and isolation, lanthanide compounds. **Actinides**-General features and chemistry of actinides, chemistry of separation of Np, Pu and Am from U, Comparison of properties of Lanthanides and Actinides and with transition elements .

UNIT-II

Second law of thermodynamics, need for the law, , Carnot's cycles and its efficiency, Carnot's theorem, .Thermodynamics scale of temperature. Concept of entropy - entropy as a state function, entropy as a function of V & T, entropy as a function of P & T,

Thermodynamics-IV Third law of thermodynamics: Nernst heat theorem, statement of concept of residual entropy, evaluation of absolute entropy from heat capacity data. Gibbs and Helmholtz functions; Gibbs function(G) and Helmholtz function (A) as thermodynamic quantities, A & G as criteria for thermodynamic equilibrium and spontaneity, their advantage over entropy change. Variation of G and A with P, V and T.

UNIT-III

Electrolytic and Galvanic cells – reversible & Irreversible cells ,conventional representation of electrochemical cells. EMF of cell and its measurement, Weston standard cell, activity and activity coefficients .Calculation of thermodynamic quantities of cell reaction (ΔG , ΔH & K). Types of reversible electrodes – metal- metal ion gas electrode, metal –insoluble salt-anion and redox electrodes. Electrode reactions, Nernst equations, derivation of cell EMF and single electrode potential. Standard Hydrogen electrode, reference electrodes, standard electrodes potential, sign conventions electrochemical series and its applications.

UNIT-IV

Structure and nomenclature of amines, physical properties. Separation of a mixture of primary, secondary and tertiary amines. Preparation of alkyl and aryl amines (reduction of nitro compounds, nitriles, reductive amination of aldehydic and ketonic compounds. Gabrielphthalimide reaction, **Diazonium Salts** Mechanism of diazotisation, structure of benzene diazoniumchloride, Replacement of diazo group by H, OH, F, Cl, Br, I, NO₂ reaction and its synthetic application.

Nitro Compounds Preparation of nitro alkanes and nitro arenes and their chemical reactions.



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UNIT-V

Nomenclature and structure of the carbonyl group. Synthesis of aldehydes and ketones with particular reference to the synthesis of aldehydes from acid chlorides, **Ketones** Mechanism of nucleophilic additions to carbonyl group with particular emphasis on benzoin, aldol, Perkin and Knoevenagel condensations. Condensation with ammonia and its derivatives. Wittig reaction. Mannich reaction. Oxidation of aldehydes, Baeyer–Villiger oxidation of ketones, Cannizzaro reaction. MPV, Clemmensen, Wolff-Kishner, LiAlH_4 and NaBH_4 reductions.

Book suggested :

1. I. L. Levine, Quantum Chemistry , Prentice-Hall Inc., New Jersey.
2. T. Engel and P. Reid, Physical Chemistry, Benjamin-Cummings.
3. S. M. Mukherji and S. P. Singh, Reaction Mechanism in Organic Chemistry, Macmillan
4. Advanced Inorganic Chemistry, F.A. Cotton and Wilkinson, John Wiley.

(Practicals)

SECTION – I (Inorganic)

Colorimetry:

1. To verify Beer - Lambert law for $\text{KMnO}_4/\text{K}_2\text{Cr}_2\text{O}_7$ and determine the concentration of the given $\text{KMnO}_4/\text{K}_2\text{Cr}_2\text{O}_7$ solution.
2. Preparations: Preparation of Cuprous chloride, prussian blue from iron fillings, tetraammine cupric sulphate, chrome alum, potassium trioxalatochromate (III).

Section-B (Physical)

3. To determine the enthalpy of solution of solid calcium chloride
- 4 .To study the distribution of iodine between water and CCl_4 .

Section-C (Organic)

Systematic identification (detection of extra elements, functional groups, determination of melting point or boiling point and preparation of at least one pure solid derivative) of the following simple mono and bifunctional organic compounds: benzoic acid, salicylic acid, aspirin, phthalic acid, cinnamic acid, benzamide, urea, acetanilide, benzanilide, aniline hydrochloride, p-toluidine, phenyl salicylate (salol), glucose, fructose, sucrose, *o*-, *m*-, *p*-nitroanilines, thiourea.

Books for practical:

1. Vogel's Textbook of Practical Organic Chemistry, A.R. Tatchell, John Wiley.
2. Practical Physical Chemistry, A.M. James and F.E. Prichard, Longman.
3. Findley's Practical Physical chemistry, B.P. Levitt, Longman.
4. Macroscale and Microscale Organic Experiments, K.L. Williamson, D.C. Heath.



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B.Sc.(Maths)

Syllabus

Semester – IV

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Mathematics)	ELECTRICITY AND MAGNETISMBASIC ELECTRONICS	BPY 141

UNIT I

Special diodes : Light Emitting Diode (LED) and its advantages – multicolor LEDs and its applications - Photo diode – characteristics and applications – Tunnel diode and its characteristics – Tunnel diode as an Oscillator – Varactor diode – Theory and its applications – Shockley diode – PIN diode and its applications.

UNIT II

Hybrid (h) parameters - determination of h -parameters – h -parameters equivalent circuit – performance of a linear circuit in h -parameter – the h -parameter of a transistor – Nomenclature for transistor h - parameters – input impedance, voltage gain and current gain in h - parameters – experimental determination of h -parameters – limitations of h -parameters.

UNIT III

Common emitter transistor as an amplifier - DC and AC load line analysis - Transistor biasing - stabilization - base resistor method - feed back resistor method - Voltage divider bias method - Construction of JFET – its characteristics and parameters - Common source JFET amplifier - MOSFET - Depletion MOSFET- Enhancement MOSFET - UJT, SCR - Construction, working, V-I characteristics and their application.

UNIT IV

Multistage amplifier – definition of gain, frequency response, decibal gain and bandwidth – operation, frequency response, advantage, disadvantage and applications of RC coupled CE transistor amplifier (two stage) and transformer coupled amplifier.

Principle of feedback in amplifiers – positive and negative feedback – effect of negative feedback - emitter follower – positive feedback amplifier as an oscillator - Hartley oscillator, Wien-bridge oscillator and Piezo electric crystal oscillator.

UNIT V

Multivibrators - astable, monostable and bistable multivibrator using transistor. Operational Amplifier : Differential amplifier - basic circuit and its operation –CMRR – Op-amp – Block diagram and explanation – applications – differentiator , integrator and comparator – multistage op-amp – solving simultaneous equations.

Books for Study:

1. Basic Electronics (Solid state), B.L. Theraja, S. Chand & Co., (2000).
2. Principles of Electronics, Metha, V.K. S. Chand & Co., (2001).



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3. Elements of electronics, M.K. Bagde and S.P. Singh, S. Chand & Co. (1982).
4. Foundations of electronics, D. Chattopadhyay and etal., New Age International Publishers (1999).
5. Hand book of Electronics - Gupta & Kumar, Pragati Prakhasan (2005).



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SCHEME

Semester – V

No	Subject Code	Subject Type	Subject Title	Marks Allotted						
				Assignment Marks		Theory Marks		Practical Marks		Total Marks
				Max	Min	Max	Min	Max	Min	
1	FC-501/1	Core	Values & Spirituality	10	4	40	14	-	-	50
2	FC-501/2	Core	Basic Computer Information Technology-I	10	4	40	14	-	-	50
3	BMM 151	Core	Real Analysis I	20	8	80	27	-	-	100
4	BCH 151	Core	Based On Inorganic, Organic And Physical Chemistry	20	8	80	27	50	17	150
5	BPY 151	Core	Nuclear Physics	20	8	80	27	50	17	150
Total				80	32	320	109	100	34	500



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B.Sc.(Maths)

B.C.A./B.A./ B.Com./B.Sc

Semester – V

Course	Subject	Subject Code
B.Sc (Mathematics)	Values & Spirituality	FC-501/1

Chapter 6: THE SUPREME- SOURCE OF VALUES

6.1	Objectives	6.2	Introduction	6.3	All about the Supreme
6.4	Name & Form of the Supreme	6.5	Abode of the Supreme		
6.6	Attributes of the Supreme	6.7	Action of benediction		
6.8	Auspicious confluence (Now or never)			6.9	Summary
6.10	Glossary	6.11	Suggested reading		

Chapter 7 : HEALING RELATIONSHIPS WITH THE SUPREME

7.1	Objectives	7.2	Introduction	7.3	Significance of the relationships
7.4	All relations with ONE	7.5	major relations	7.6	True and eternal relations
7.7	Benefits of various relationships	7.8	The timeless dimensions	7.9	Summary
7.10	Glossary	7.11	Suggested reading		

Chapter 8 : RAJYOGA MEDITATION

8.1	Objectives	8.2	Introduction	8.3	Methods of Meditation
8.4	Rajyoga meditation with a difference	8.5	Five fold impact of Rajyoga meditation		
8.6	Stages of Rajyoga meditation	8.7	Attainments of Meditation		
8.8	Research studies on meditation	8.9	Summary		
8.10	Glossary	8.11	Suggested reading		

Chapter 9 : SPIRITUAL LIFE STYLE

9.1	Objectives	9.2	Introduction	9.3	Early morning meditation
9.4	Regular spiritual study	9.5	Authentic life style	9.6	Satwic Diet
9.7	Selfless service	9.8	Review of Personal Progress		
9.9	Summary	9.10	Glossary	9.11	Suggested reading

Chapter 10 : EXERCISES FOR PRACTICE

Quiet reflection- Practice introversion-Being an observer-Stand back and observe -Self awareness (Soul consciousness)-Experiencing Body free stage-Reflect on original qualities-Visualize the Divine-Think attributes of the Supreme-Developing a living relationship-Surrender to God-Create Good wishes for all-Visualization in Meditation: Orbs of Light- The forest-The Balloon



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B.C.A./B.A./ B.Com./B.Sc

Semester – V

Course	Subject	Subject Code
B.Sc (Mathematics)	Basic Computer Information Technology-I	FC-501/2

Unit – I

INTRODUCTION TO COMPUTER ORGANIZATION –I :- History of development of Computer system concepts. Characteristics, Capability and limitations. Generation of computer. Types of PC's Desktop, Laptop, Notebook. Workstation & their Characteristics.

कम्प्यूटर ऑर्गनाइजेशन का परिचय – I

कम्प्यूटर का इतिहास, कम्प्यूटर सिस्टम विचारधारा, विशेषताएं, योग्यता एवं सीमाएं, कम्प्यूटर की पीढ़ियां, पी.सी. के प्रकार, डेस्कटॉप के प्रकार, लेपटॉप के प्रकार, नोटबुक, वर्कस्टेशन आदि की विशेषताएं।

Unit – II

INTRODUCTION TO COMPUTER ORGANIZATION –II :- Basic components of a computer system Control Unit, ALU. Input/ Output function and Characteristics, memory RAM, ROM, EPROM, PROM.

कम्प्यूटर ऑर्गनाइजेशन का परिचय – II

कम्प्यूटर सिस्टम के आधार उपकरण, कंट्रोल युनिट, ए.एल.यू. इनपूट/ आउटपुट फंक्शन और विशेषताएं, मेमोरी रेम, रोम, ईपी रोम, पी रोम, और अन्य प्रकार की मेमोरी।

Unit – III

INPUT & OUTPUT DEVICES :- Input Devices : Keyboard, Mouse, Trackball. Joystick, Digitizing tablet, Scanners, Digital Camera, MICR, OCR, OMR, Bar-code Reader, Voice Recognition, Light pen, Touch Screen. **Output Devices:** Monitors Characteristics and types of monitor, Video Standard VGA, SVGA, XGA, LCD Screen etc. Printer, Daisy wheel, Dot Matrix, Inkjet, Laser, Line Printer. Plotter, Sound Card and Speakers.

इनपुट तथा आउटपुट डिवाइसेस

इनपुट डिवाइस : कीबोर्ड, माउस, ट्रैकबॉल, जॉयस्टिक, डिजिटाइजिंग टेबलेट, स्कैनर्स, डिजिटल केमरा, एमआईसीआर, ओसीआर, ओएमआर, बार कोड रीडर, आवाज को पहचानने वाला, लाइट पेन, टच स्क्रीन।

आउटपुट डिवाइस : मॉनीटर की विशेषताएं एवं मॉनीटर के प्रकार, वीडियो स्टैंडर्ड VGA, SVGA, XGA, LCD स्क्रीन आदि, प्रिंटेर्स, डेजी व्हील, डॉट मैट्रिक्स, इंकजेट, लेजर, लाइन प्रिंटर, प्लॉटर, साउंड कार्ड्स एवं स्पीकर्स।

Unit – IV

STORAGE DEVICES :- Storage fundamental primary Vs Secondary. Various Storage Devices magnetic Tape. Cartridge Tape, Data Drives, Hard Drives, Floppy Disks, CD, VCD, CD-R, CD-RW, Zip Drive, DVD, DVD-RW.

स्टोरेज डिवाइजेस

स्टोरेज फंडामेंटल्स प्राइमरी विरुद्ध भिन्न स्टोरेज डिवाइजेस मेग्नेटिक टेप, कार्ट्रिज टेप, डाटा ड्राइव्स, हार्ड डिस्क ड्राइव्स, फ्लोपी डिस्कस, सी.डी. वी.सी.डी. सी.डी.-आर.सी.डी.- आर. डब्ल्यू, जीप ड्राइव, डी.वी.डी., डी.वी.डी., - आर. डब्ल्यू।



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Unit – V

INTRODUCTION TO OPERATING SYSTEM :- Introduction to operating systems, its functioning and types. basic commands of dos & Windows operating System.

Disk Operating System (DOS)

- Introduction, History and Versions of DOS.

DOS Basics

- Physical Structure of disk, Drive name, FAT, file & directory structure and naming rules, booting process, DOS system files.

DOS Commands

- Internal DIR, MD, CD, RD, Copy, DEL, REN, VOL, DATE, TIME, CLS, PATH, TYPE etc.
- External CHKDSK, SCOPE, PRINT DISKCOPY, DOSKEY, TREE, MOVE, LABEL, APPEND, FORMAT, SORT, FDISK, BACKUP, MODE, ATTRIB HELP, SYS etc.

ऑपरेटिंग सिस्टम का परिचय

ऑपरेटिंग सिस्टम का परिचय, उसके लक्षण एवं प्रकार, डॉस एवं विन्डोज का मूल कमांड। ऑपरेटिंग सिस्टम : डिस्क ऑपरेटिंग सिस्टम (DOS), परिचय इतिहास एवं वर्जन्स ऑफ डॉस।

डॉस बेसिक्स –

फिजीकल स्ट्रक्चर ऑफ डिस्क, ड्राइव नेम, फेट, फाईल एवं डायरेक्ट्री स्ट्रक्चर एवं नेमिंग नियम, बूटिंग प्रक्रिया, डॉस सिस्टम फाईल्स।

डॉस कमाण्डस–

- आंतरिक कमाण्डस DIR, MD, CD, RD, Copy, DEL, REN, VOL, DATE, TIME, CLS, PATH, TYPE आदि।
- बाह्य कमाण्डस CHKDSK, SCOPE, PRINT DISKCOPY, DOSKEY, TREE, MOVE, LABEL, APPEND, FORMAT, SORT, FDISK, BACKUP, MODE, ATTRIB HELP, SYS आदि।

Books Recommended-

- 1- डॉ. एस. के. विजय, डॉ. पंकज सिंह : कम्प्यूटर विज्ञान एवं सूचना प्रौद्योगिकी, मध्यप्रदेश हिन्दी ग्रन्थ अकादमी, भोपाल
- 2- डॉ. पंकज सिंह कम्प्यूटर अध्ययन, राम प्रसाद एड संस

Course	Subject	Subject Code
B.Sc (Mathematics)	Basic Computer Information Technology-I	FC 501/2

PRACTICALS

DOS :

- DOS commands : Internal & External Commands.
- Special batch file : Autoexec, Bar Hard disk setup.

Windows 98:

- Desktop setting : New folder, rename bin operation, briefcase, function. Control panel utility.
- Display properties: Screen saver, background settings.

Ms-Word:



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- Creating file: save, save as HTML, Save as Text, template, RTF Format.
- Page setup utility: Margin settings, paper size setting, paper source, layout.
- Editing: Cut, paste special, undo, redo, find, replace, goto etc.
- View file: page layout, Normal Outline, master document, ruler header, footer, footnote, full screen.
- Insert: break, page number, symbol, date & time, auto text, caption file, object, hyperlink, picture etc.
- Format: font, paragraph, bullets & numbering, border & shading, change case, columns.
- Table : Draw label, insert table, cell handling, table auto format, sort formula.



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B.Sc.(Maths)

Semester – V

Branch	Subject title	Subject code
B.Sc (Mathematics)	Real Analysis - I	BMM-151

UNIT:-I

Sets and functions :-Sets and elements; operations on sets; functions; real valued functions; equivalence; countability; real numbers; least upper bounds.

Sequences of Real Numbers :-Definition of a sequence and subsequence; limit of a sequence; convergent sequences; divergent sequences; bounded sequences; monotone sequences; operations on convergent sequences; operations on divergent sequences; limit superior and limit inferior; Cauchy sequences.

UNIT:-II

Series of Real Numbers :-Convergence and divergence; series with non-negative numbers; alternating series; conditional convergence and absolute convergence; tests for absolute convergence; series whose terms form a non-increasing sequence; the class 1^2 .

Limits and metric spaces :-Limit of a function on a real line; metric spaces; limits in metric spaces.

UNIT:-III

Approximations, Errors and its types, Solution of Equations: Bisection, Secant, Regula Falsi, Newton- Raphson Method and their order of convergence, Roots of second degree Polynomials, Interpolation: Lagrange interpolation, Divided Differences, Interpolation formulae using Differences and derivations of Interpolation formula..

UNIT:-IV

Linear Equations: Direct Methods for Solving Systems of Linear Equations, Gauss elimination, Gauss Jordan Method, LU Decomposition, Cholesky Decomposition, Iterative Methods: Jacobi Method , Gauss - Seidel Method, Relaxation Method, Methods Based on Numerical Differentiation..

UNIT:-V

Ordinary Differential Equations: Euler Method, Eulers Modified Method, Single-step Methods, Runge-Kutta's Method, Multi-step Methods, Milne Method, Numerical Quadrature, Newton-Cote's Formulae, Gauss Quadrature Formulae, Methods Based on Numerical Integration with their derivation.

Reference Book :-1. Treatment as in "Methods of Real Analysis" : Richard. R. Goldberg (Oxford and IBH Pubhshing Co.)



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2. C E Frooerg. Introduction to Numerical Analysis, (Second Edition L Addison-Wesley - 1979,
3. M K Jain, S.R.K. Iyengar, R. K. Jain. Numerical Methods Problems and Solutions, New Age International (P)Ltd. 1996.
- 4 E. Balaguruswamy- Numerical Method Tata Mc Graw_ Hill Pub.Com. New Yark 5
- K.B. Datta. Matrix and Linear Algebra, Prentice hall of India Pvt Ltd., New Delhi, 2000.



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B.Sc.(Maths)

Syllabus Semester – V

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Mathematics)	BASED ON INORGANIC, ORGANIC & PHYSICAL CHEMISTRY	BCH-151

UNIT-I

Limitations of valence bond theory, an elementary idea of crystal-field theory, crystal field splitting in octahedral, tetrahedral and square planar complexes, **Magnetic Properties of Transition Metal Complex** Types of magnetic behavior, methods of determining magnetic susceptibility, spin-only formula. L-S coupling, **Electron Spectra of Transition Metal Complexes** Types of electronic transitions, selection rules for d-d transitions, spectroscopic ground states, spectrochemical series. Orgel-energy level diagram for d1 and d9 states, discussion of the electronic spectrum of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ complex ion.

UNIT-II

Black-body radiation, Planck's radiation law, photoelectric effect, heat capacity of solids, Compton effect, wave function and its significance of Postulates of quantum mechanics, quantum mechanical operator, commutation relations, Hamiltonian operator, Hermitian operator, average value of square of Hermitian as apposite quantity, Role of operators in quantum mechanics, To show quantum mechanically that position and momentum cannot be predicated simultaneously, Determination of wave function & energy of a particle in one dimensional box, Pictorial representation and its significance.

UNIT-III

Introduction: Electromagnetic radiation, regions of spectrum, basic features of spectroscopy, **Rotational Spectrum** Diatomic molecules. Energy levels of rigid rotator (semi-classical principles), selection rules. **Vibrational spectrum** Infrared spectrum: Energy levels of simple harmonic oscillator, selection rules, pure vibrational spectrum, effects of an harmonic motion and isotopic effect on the spectra., idea of vibrational frequencies of different functional groups. **Raman Spectrum:** Concept of polarizability, pure rotational and pure vibrational Raman spectra of diatomic molecules, selection rules, Quantum theory of Raman spectra.

UNIT-IV

Classification and nomenclature. Monosaccharides, mechanism of osazone formation, inter conversion of glucose and fructose, chain lengthening and chain shortening of aldoses. Configuration of monosaccharides. Erythro and threo diastereomers. Conversion of glucose in to mannose. Formation of glycosides, ethers and esters. Determination of ring size of glucose and fructose. Open chain and cyclic structure of D(+)-glucose & D(-) fructose. Mechanism of mutarotation. Structures of ribose and deoxyribose. An introduction to disaccharides (maltose, sucrose and lactose) and polysaccharides (starch and cellulose) without involving structure determination.



RKDF UNIVERSITY, BHOPAL

B.Sc.(Maths)

UNIT-V

Organomagnesium compounds: the Grignard reagents-formation, structure and chemical reactions. Organozinc compounds: formation and chemical reactions. Organolithium compounds: formation and chemical reactions.

Books:

1. P. S. Kalsi., Organic Reactions and their Mechanisms, New Age International
2. Chemistry of the Elements. N.N. Greenwood and A. Earnshaw, Pergamon.
3. Chemistry of the Elements. N.N. Greenwood and A. Earnshaw, Pergamon.
4. P.W. Atkins , Physical Chemistry, ELBS.

(Practical)

SECTION – I (Inorganic)

Semimicro qualitative analysis of mixture containing not more than four radicals (including interfering, Combinations and excluding insolubles):

Pb²⁺, Hg²⁺, Hg₂²⁺, Ag⁺, Bi³⁺, Cu²⁺, Cd²⁺, As³⁺, Sb³⁺, Sn²⁺, Fe³⁺, Cr³⁺, Al³⁺, Co²⁺, Ni²⁺, Mn²⁺, Zn²⁺, Ba²⁺, Sr²⁺, Ca²⁺, Mg²⁺, NH₄⁺, CO₃²⁻, S²⁻, SO₃²⁻, S₂O₃²⁻, NO₂⁻, CH₃COO⁻, Cl⁻, Br⁻, I⁻, NO₃⁻, SO₄²⁻, C₂O₄²⁻, PO₄³⁻, BO₃

Section-B (Physical)

1. To determine the strength of the given acid solution (mono and dibasic acid) conductometrically.
2. To determine the solubility and solubility product of a sparingly soluble electrolyte conductometrically
3. To determine the strength of given acid solution (mono and dibasic acid)/KMnO₄ – Mohr salt potentiometrically.

Section-C (Organic)

1. Laboratory Techniques

(a) **Steam distillation** (non evaluative) Naphthalene from its suspension in water Separation of o- and p-nitrophenols

(b) **Column chromatography** (non evaluative)

Separation of fluorescein and methylene blue Separation of leaf pigments from spinach leaves

2. Chromatography Method

Determination of R_f values and identification of organic compounds

(a) Separation of green leaf pigments (spinach leaves may be used) by paper chromatographic method

(b) Separation of a mixture of coloured organic compounds using common organic solvents by TLC.

3. Synthesis of the following organic compounds:

1. To prepare p-bromoaniline from p-bromoacetanilide.
2. To prepare m-nitroaniline from m-dinitrobenzene.



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B.Sc.(Maths)

Syllabus Semester – V

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Mathematics)	NUCLEAR PHYSICS	BPY 151

UNIT I

RADIOACTIVITY:

Laws of successive disintegration - transient - and secular equilibria - range of alpha particles - experimental measurement - Geiger - Nuttall Law - alpha ray spectra - Gamow's theory of alpha decay and its experimental verification - Beta ray spectra - origin of line and continuous spectrum - Fermi's theory of beta decay - K electron capture - Nuclear Isomerism.

UNIT II

NUCLEAR DETECTORS:

Principle and working - solid state detector - proportional counter - Wilson's cloud chamber - Scintillation counter.

Accelerators : Synchrocyclotron - Synchrotron - Electron synchrotron - proton synchrotron - Betatron.

UNIT III

ARTIFICIAL TRANSMUTATION:

Rutherford's experiment - Bohr's theory of Nuclear disintegration – Q value equation for a nuclear reaction - threshold energy - types of nuclear reaction - energy balance and the Q value - threshold energy of an endoergic reaction.

Neutron: Mass, charge, decay, spin and magnetic moment, Neutron diffraction, absorption of neutron by matter - neutron sources - detectors - neutron collimator.

UNIT IV

NUCLEAR STRUCTURE: General properties of nucleus – size , mass and charge.

Proton - electron theory - proton - neutron theory - Nuclear size - experimental measurement of nuclear radius - mirror nuclei method - meson theory of nuclear forces - nuclear models - liquid drop model - Weizacker's semi - empirical formula - nuclear shell model.

UNIT V

NUCLEAR FISSION, FUSION AND ELEMENTARY PARTICLES

Nuclear fission - Bohr Wheeler theory - chain reaction – critical size and critical mass - Nuclear fission reactor – Nuclear fusion - source of stellar energy - Carbon – Nitrogen cycle - Proton – Proton cycle - Thermo Nuclear reaction – plasma.

Elementary Particles - types of interactions - classification of elementary particles - particle quantum numbers – baryon number - lepton number - strangeness number - hyper charge - spin quantum number.



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B.Sc.(Maths)

Books for Study:

1. Modern Physics by R. Murugesan, S.Chand & Co., (2005)
2. Atomic Physics by J.B. Rajam, S.Chand & Co., (2005)
3. Nuclear Physics by D.C.Tayal, Himalaya Publishing House.

Books for Reference:

1. A Source book on Atomic energy by Samuel Glass Stone (2002)
2. Atomic and Nuclear Physics by Albright Semat (2003)
3. Atomic and Nuclear Physics by Little field and Thorley. ELBS (2002)
4. Basic Nuclear Physics and Cosmic rays, B.N. Srivatsava, Pragti Prakasham.



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B.Sc.(Maths)

SCHEME

Semester – VI

No	Subject Code	Subject Type	Subject Title	Marks Allotted						
				Assignment Marks		Theory Marks		Practical Marks		Total Marks
				Max	Min	Max	Min	Max	Min	
1	FC-601/1	Core	Values & Spirituality	10	4	40	14	-	-	50
2	FC-601/2	Core	Basic Computer Information Technology-I	10	4	40	14	-	-	50
3	BMM 161	Core	Programmming Language with 'C' Theory	20	8	80	27	-	-	100
4	BCH 161	Core	Based On Inorganic, Organic And Physical Chemistry	20	8	80	27	50	17	150
5	BPY 161	Core	Electronic Communication	20	8	80	27	50	17	150
Total				80	32	320	109	100	34	500



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B.Sc.(Maths)

B.C.A./B.A./B.Com./B.Sc

Semester – VI

Course	Subject	Subject Code
B.Sc (Mathematics)	भाषा कौशल एवं व्यक्तित्व विकास	FC-601/1

इकाई –1

- | | |
|--|-------------------------|
| 1. भारतीय संस्कृति | 2. भारतीय समाज व्यवस्था |
| 3. सभ्यता एवं संस्कार | 4. वैश्विक चेतना |
| 5. समन्वयीकरण (भारतीय एवं अंतर्राष्ट्रीय संदर्भ में) | |

bdkb & 2

- | | | |
|---------|------------|----------|
| 1. धर्म | 2. न्याय | 3. दर्शन |
| 4. नीति | 5. साहित्य | |

bdkb & 3

- | | |
|---|----------------|
| 1. संचार संसाधन : सम्पर्क के नए क्षितिज | 2. समाचार पत्र |
| 3. भारतीय प्रेस परिषद् | 4. रेडियो |
| 5. दूरदर्शन | |

bdkb & 4

- | | | |
|--------------------------------|--------------|----------|
| 1. सिनेमा | 2. रंगमंच | 3. संगीत |
| 4. चित्र, मूर्ति, स्थापत्य कला | 5. शिल्प कला | |

bdkb & 5

- | | |
|--|---------------------------------------|
| 1. कम्प्यूटर | 2. दूरभाष : विज्ञान की सौगात |
| 3. मंत्र (कहानी) : प्रेमचंद | 4. मातृभूमि (कविता) : मैथिलीशरण गुप्त |
| 6. साहित्यकार का दायित्व : डॉ. प्रेम भारती | |

1. नमो भगवते वासुदेवाय



RKDF UNIVERSITY, BHOPAL

B.Sc.(Maths)

B.C.A./B.A./B.Com./B.Sc

Semester – VI

Course	Subject	Subject Code
B.Sc (Mathematics)	Basic Computer Information Technology-II	FC-601/2

Unit-I

Word Processing: Word

- Introduction to word Processing.
- MS Word: features, Creating, Saving and Operating Multi document windows, Editing Text selecting, Inserting, deleting moving text.
- Previewing documents, Printing document to file page. Reduce the number of pages by one.
- Formatting Documents: paragraph formats, aligning Text and Paragraph, Borders and shading, Headers and Footers, Multiple Columns.

Unit-II

Introduction to Excel

Excel & Worksheet:

- Worksheet basic.
- Creating worksheet, entering data into worksheet, heading information, data text, dates, alphanumeric, values, saving & quitting worksheet.
- Opening and moving around in an existing worksheet.
- Toolbars and Menus, keyboard shortcuts.
- Working with single and multiple workbook coping, renaming, moving, adding and deleting. coping entries and moving between workbooks.
- Working with formulas & cell referencing.
- Autosum.
- Coping formulas
- Absolute & Relative addressing.

Unit-III

Introduction to Power Point

- Features and various versions.
- Creating presentation using Slide master and template in various colour scheme.
- Working with slides make new slide move, copy, delete, duplicate, lay outing of slide, zoom in or out of a slide.
- Editing and formatting text: Alignment, editing, inserting, deleting, selecting, formatting of text, find and replace text.

Unit-IV



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B.Sc.(Maths)

Power Point– II

- Bullets , footer, paragraph formatting, spell checking.
- Printing presentation Print slides, notes, handouts and outlines.
- Inserting objects Drawing and Inserting objects using Clip Arts picture and charts.
- Slide sorter, slide transition effect and animation effects. Presenting the show making stand alone presentation, Pack and go wizards.

Unit – V

Evolution, Protocol, concept, Internet, Dial-up connectivity, leased line, VSAT, Broad band, URLs, Domain names, Portals. E-mail, Pop & web based Email. Basic of sending and receiving Emails, Email & Internet Ethics, Computer virus, Antivirus software wage, Web Browsers.

Books Recommended-

1. डॉ. एस. के. विजय, डॉ. पंकज सिंह : कम्प्यूटर विज्ञान एवं सूचना प्रौद्योगिकी, मध्यप्रदेश हिन्दी ग्रन्थ अकादमी, भोपाल
2. डॉ. पंकज सिंह कम्प्यूटर अध्ययन, राम प्रसाद एड संस

Course	Subject	Subject Code
B.Sc (Mathematics)	Basic Computer Information Technology-I	FC 601/2

Practical

Ms-Power Point:

Creating new slide, formatting slide layout, slide show & sorter, Inserting new slide, slide no., date, time, chart, formatting slide, tool operation.

List of suggested practical work:

- Under standing of a dial up connection through modern.
- Configuring a computer for an e-mail and using outlook Express or Netscape Messenger.
- Registration an e-mail address.
- Understanding of e-mail drafting.
- Understanding of address book maintenance for e-mail.
- Understanding of different mail program tools.
- Send and receive functions of e-mail.



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B.Sc.(Maths)

Semester – VI

Branch	Subject title	Subject code
B.Sc (Mathematics)	Programming Language with “C” Theory	BMM-161

UNIT:-I

C constants, Variables, Data-types, Declaration of variables, Assigning values to variables.

Operators: Arithmetic, Relational, Logical, Increment and Decrement Conditional Operators. Arithmetic Expressions, Evaluation of Expressions, Precedence of arithmetic operators, Formatted Input and Output, Decision making and branching, If, Simple-If, If-Else, Nesting of If-Else-If Ladder, Switch Statement, The ?: Operator, Go to Statement

Decision making with looping: While, Do, For Statement, Jumps in loops
Arrays: 1-dimensional, 2-dimensional array, Initialising 2-D arrays, Multi-Dimensional arrays

UNIT:-II

User-Defined Function: Need for user-defined function, Multi-Function Program, The form of C-Function return value and their types, Calling a function, Category of functions, No arguments and no return values, arguments but no return values, arguments with return values. Handling of non-integer functions, Nesting of functions, Recursion, Function with Arrays, Scope and lifetime of variables in functions
Structure and Unions: Structure definition, Giving values to numbers, Structure initialisation, Comparison of structure variables, Arrays of structure, Arrays within Structures, Structure within Structures, Structures and functions.

UNIT:-III

Pointers: Understanding pointers, accessing the address of a variable, Declaring and initialising of pointers, accessing a variable through its pointer, Pointer expression, Pointer increments and scale factor, Pointers and arrays, Pointers and functions, Pointers and structures
Files: Defining and opening a file, closing a file.

UNIT:-IV

Boolean Function, Disjunction and Conjunction Normal Forms, Boole's Expansion, Theorem. Binary Relations, Equivalence Relations, Partitions and Partial Order Relation.

UNIT:-V



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B.Sc.(Maths)

Graphs, Multigraphs, Weighted Graphs, Paths and Circuits, Shortest Paths: Dijkstra's Algorithm, Matrix Representation of Graph: Incidence and Adjacency Matrix, Trees and its simple properties.

PRACTICAL LIST OF PROGRAMME:-

- 1.The Factorial of an Integer
2. Binomial coefficient nCr , where n and r are Positive Integers
3. Matrix addition and subtraction.
4. Matrix multiplication.
5. Transpose of a Matrix.
6. Insertion sort
7. Bisection method
8. Solve a System of Linear Simultaneous equations by Gauss-Elimination method
- 9.To evaluate a definite integral using Simpson's one-third rule.
- 10.To solve an Ordinary Differential equation of first order by Runge-Kutta method of fourth order

Text Books :

1. R.R Goldberg, Real Analysis, Oxford & IBH Publishing Co., New Delhi, 1970.
2. G.F. Simmons. Introduction to Topology and Modern Analysis. McGraw-Hill, 1963.
3. T.M Apostol, Mathematical Analysis. Narosa Publishing House. New Delhi, 1
4. C.L. Liu, Elements of Discrete Mathematics, (Second Edition), McGraw Hill, International Edition, Computer Science series 1986.

Reference Books:

- 1 Programming in ANSI C 2nd edition, E.Balaguruswamy, Tata-Mcgraw Hill Publishing company.
2. T.M Apostol, Mathematical Analysis. Narosa Publishing House. New Delhi, 1985.
3. S. Lang. Undergraduate Analysis, Springer-Verlag, New York, 1983.
4. D. Somasundaram and B. Choudhary, A first Course in Mathematical Analysis. Narosa Publishing House, New Delhi 1997.
5. Shanti Narayan, A Course of Mathematical Analysis. S. Chand & Co. Delhi.
6. RK. Jain and S.K. Kaushik, An introduction to Real Analysis, S. Chand & Co., New Delhi 2000.



RKDF UNIVERSITY, BHOPAL

B.Sc.(Maths)

Syllabus
Semester – VI

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Mathematics)	Based on inorganic, Organic & Physical Chemistry	BCH-161

UNIT-I

Arrhenius, Bronsted – Lowry, the Lux – Flood, Solvent system and Lewis concepts of acids & bases, relative strength of acids & bases, Concept of Hard and Soft Acids & Bases. Symbiosis, electronegativity and hardness and softness.

Bioinorganic Chemistry

Essential and trace elements in biological processes, metalloporphyrins with special reference to haemoglobin and myoglobin. Biological role of alkali and alkaline earth metal ions with special reference to Ca^{2+} . Nitrogen fixation.

UNIT-II

Interaction of radiation with matter, difference between thermal and photochemical processes. Laws of photochemistry: Grotthus-Draper law, Stark-Einstein law (law of photochemical equivalence) Jablonski diagram depicting various processes occurring in the excited state, qualitative description of fluorescence, phosphorescence, non-radiative processes (internal conversion, intersystem crossing), quantum yield, photosensitized reactions-energy transfer processes (simple examples).

UNIT-III

Statement and meaning of the terms – phase component and degree of freedom, thermodynamic derivation of Gibbs phase rule, phase equilibria of one component system – Example – water and Sulphur systems. Phase equilibria of two component systems solid-liquid equilibria, simple eutectic Example Pb-Ag system, desilverisation of lead.

UNIT-IV

Introduction: Molecular orbital picture and aromatic characteristics of pyrrole, furan, thiophene and pyridine. Methods of synthesis and chemical reactions with particular emphasis on the mechanism of electrophilic substitution. Mechanism of nucleophilic substitution reactions in pyridine derivatives. **Heterocyclic Compounds-II** Introduction to condensed five and six- membered heterocyclic. Preparation and reactions of indole, quinoline and isoquinoline with special reference to Fisher indole synthesis, Skraup synthesis and Bischler-Napieralski synthesis.



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B.Sc.(Maths)

UNIT-V

Nomenclature, structural features, Methods of formation and chemical reactions of thiols, thioethers, sulphonic acids, sulphonamides and sulphaguanidine. Synthetic detergents alkyl and aryl sulphonates. **Amino Acids, Peptides & Proteins**

Classification, of amino acids. Acid-base behavior, isoelectric point and electrophoresis. Preparation of β -amino acids. Structure and nomenclature of peptides and proteins. Classification of proteins. Peptide structure determination, end group analysis, selective hydrolysis of peptides. Classical peptide synthesis, solid-phase peptide synthesis. Structures of peptides and proteins: Primary & Secondary structure.

Books:

1. Ira N. Levine, Quantum Chemistry, Prentice Hall.
2. F. A. Carey and R. J Sundberg, Advanced Organic Chemistry, Part A and B, Plenum
3. K.J. Laidler, Chemical Kinetics. McGraw-Hill.
4. Inorganic Chemistry, J.E. Huhey, Harpes & Row.
5. Valence, C. A. Coulson, Oxford University Press.

(Practical)

Section-B (Physical)

1. To determine the molecular weight of a non-volatile solute by Rast method.
2. To standardize the given acid solution (mono and dibasic acid) pH metrically.

Section-C (Organic)

2. Chromatography Method

Determination of R_f values and identification of organic compounds

(a) Separation of green leaf pigments (spinach leaves may be used) by paper chromatographic method

(b) Separation of a mixture of coloured organic compounds using common organic solvents by TLC.

3. Synthesis of the following organic compounds:

1. To prepare o-chlorobenzoic acid from anthranilic acid.
2. To prepare S-Benzyl-iso-thiourea from thiourea.

Book for practical:

1. Synthesis and Characterization of Inorganic Compounds, W.L. Jolly. Prentice Hall.
2. Experiments and Techniques in Organic Chemistry, D.P. Pasto, C. Johnson and M. Miller, Prentice Hall.
3. Practical Physical Chemistry, A.M. James and F.E. Prichard, Longman.



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B.Sc.(Maths)

Syllabus

Semester – VI

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Mathematics)	ELECTRONIC COMMUNICATION	BPY 161

UNIT I

Modulation - definition - types of modulation AM, FM, PM – expression for amplitude modulated voltage - wave form of amplitude modulated wave - collector modulation circuit - single side band generation - balanced modulator - AM transmitter - block diagram and explanation - frequency modulation - expression for frequency modulated voltage - side bands in FM, AM production by transistor modulator - comparison of AM, FM, PM.

UNIT II

Demodulation - definition - diode detection of AM signals - FM detection - Foster Seely discriminator - radio receivers - straight receivers - TRF receivers - super heterodyne receivers - block diagram - explanation of each stage - FM receivers – block diagram.

UNIT III

TV - plumbicon - vidicon - scanning and interlaced scanning – block diagram of TV transmitter and receiver - colour TV - generation R, G, B signals - simplified block diagram of colour TV transmitter and receiver – TV transmitting antennas - dipole panel - TV receiving antenna - Yagi antenna - log antenna - log periodic antenna.

UNIT IV

RADAR - principle of radar - Radar equation - radar – transmitting systems - radar antennas - duplexer - radar receivers uses of radar – opto-electronic devices – photoconductive cell - solar cell - phototransistor – LED and LCD - construction and working.

UNIT V

Digital communications - digital technology - fundamentals of data communication systems - characteristics of data transmission circuits – digital codes - error detection and correction - data sets and inter connection - requirements - modern classification - modern interfacing.

Books for Study:

1. Hand book of Electronics - Gupta & Kumar, Pragati Prakhasan (2005).
2. Electronics Communication Systems - Kennedy and Davis, TMH.
3. Electronics Communications – Roody Coolen, Pearson Education Publication.
4. Electronics Communications - Frenzel, TMH.